

Let's take a closer look at what this means for enterprises and IT workers.

Lollipop aims to provide a physical separation between a user's personal data and enterprise data, using the Knox framework. There will be a separate launcher app for enterprise apps, and all enterprise applications will be marked by the 'Android for Work' icon.

This separation will empower IT administrators to control enterprise apps and their corresponding data without affecting the user's personal applications and data. All leading EMM providers are claiming support of Lollipop by leveraging the native capabilities offered by Google.

Figure 1 gives a comparative view of the 'pre-Lollipop' and 'Lollipop' eras.

Changes at the application layer and the Android OS layer are expected to bring enterprise-class security in a default manner in Android Lollipop-powered devices. All EMM providers will then be able to leverage this default support in their upcoming releases.

A typical EMM solution consists of the following:

- A web console
- An on-device EMM agent
- An EMM library to be used with enterprise applications for policy compliance

In the pre-Lollipop scenario, an on-device EMM agent and the EMM library were specific to particular EMM providers. Hence, the enterprise acceptability of an Android device was driven by the capabilities of the selected third-party EMM solution.

But now, in the Lollipop era, the equivalent of an on-device agent will be provided by Google, and 'Android for Work' APIs will replace the EMM library. Hence, going forward, it can be assumed that, by default, Lollipop-powered devices will be enterprise ready. However, third party EMM solutions will still be required to perform administrative activities like enterprise policies management, their deployment on selected sets of devices, etc., in a remote manner.

Figure 3 shows an overview of a typical EMM solution in the Lollipop era.

Device and data security

We have already discussed the strong separation between personal and enterprise data and applications in Lollipop. Apart from this, Lollipop includes a device protection feature called the Kill Switch. If this anti-theft protection is enabled on the device and the 'Lock password' feature is available, then the device's 'Factory reset' option will ask the user for their registered Google ID credentials. Without providing valid credentials, a thief will not be able to implement a factory reset on a stolen device, and the device will remain unusable.



Figure 1: Comparative view

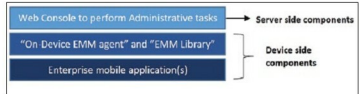


Figure 2: A typical EMM solution

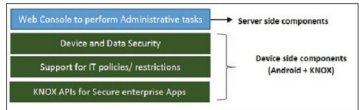


Figure 3: An EMM solution in the era of Android Lollipop

Support for IT policies and restrictions

EMM providers will be able to use a new set of APIs powered by Knox to enforce a wide set of policies ranging from system settings to application-specific settings.

Knox APIs for secure enterprise apps

EMM providers will be able to use new backend APIs powered by Knox APIs. This will empower IT administrators to take care of remotely deployed devices and securely manage the applications installed on them.

Android Lollipop is the largest and most ambitious release from Google, especially from an enterprise mobility perspective. Last but not least, all the changes discussed above will enable Google to address OEM fragmentation issues to a certain extent. **END** 

References

- [1] <http://android-developers.blogspot.in/2014/07/knox-contribution-to-android.html>
- [2] <https://www.samsungknox.com/en/androidworkwithknox>

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